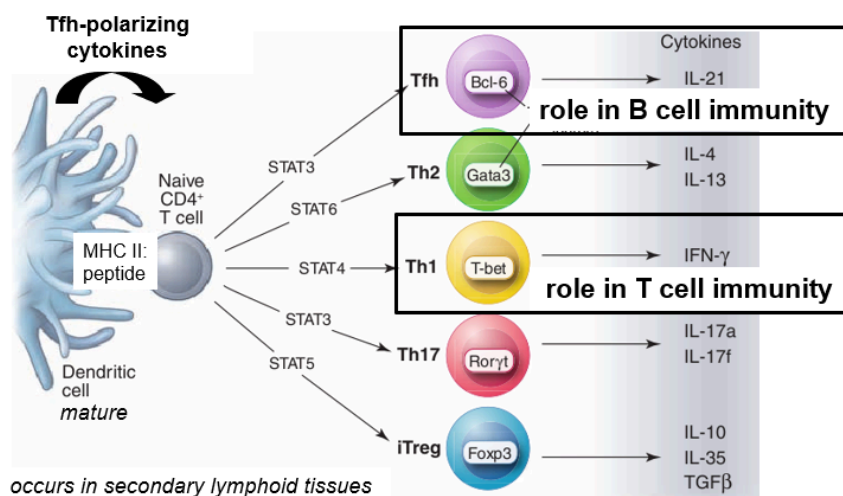


Topic 7-2

- **Effective helper (Th) T cells** help other cells by “telling” them how to eliminate pathogens via release of **cytokines**
 - Dn destroy pathogens directly
- Many diff types of Effector Th cells
 - For a naive T cell to know which type of helper cell to become - all dictated by the mature dendritic cell (DC)
 - Mature DC 1) bind to naive T helper cells and 2) secrete polarizing cytokines telling that naive T helper cell to become (in this case) an effector T helper 1 cell (yellow one)

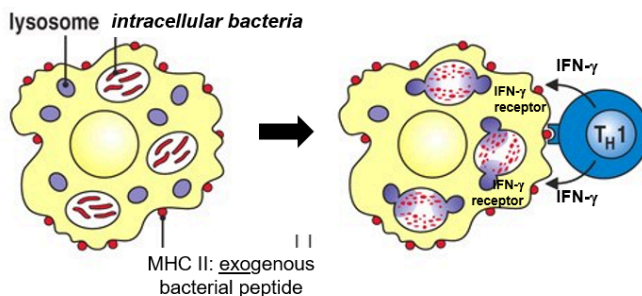
Different Types of Effector Th cells



- Role of **Effector T Helper 1 (Th1) cells** is to eliminate intracellular bacteria that are living in the **phagosome/phagolysosome** of **macrophages**
 - (Recall: those intracellular bacteria that are in the cytoplasm of the phagocyte will be destroyed by the effector cytotoxic T cells)
 - Effector Th1 cells are ineffective here
- Why are macrophages predominantly the cell infected by intracellular bacteria?
 - Bc many intracellular bacteria require iron for growth/survival in host cells
 - Macrophages are important in normal iron homeostasis - they phagocytose RBC (dn need to know for exam)
- Some intracellular bacteria are still destroyed by lysosomal contents - so there is always gonna be some exogenous bacterial peptide being presented on MHC class 2 molecules by the infected macrophage

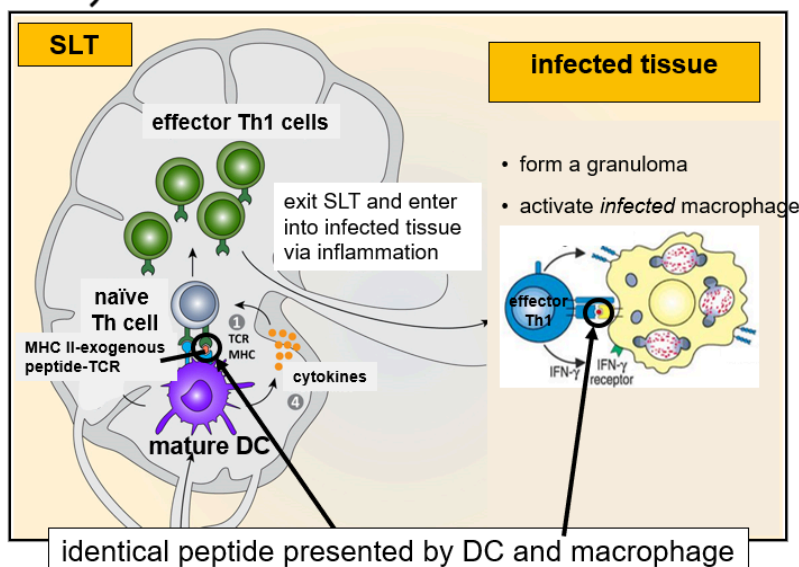
- Effector Th1 cells bind to peptide on the MHC class 2 molec, and secrete cytokine: IFN- γ (gamma)
- IFN- γ will bind to interferon gamma receptor present on the surface of the macrophage
 - this sends activation signal to the infected macrophage to tell it to increase lysosome-phagosome fusion -> more acids and lysosomal enzyme released onto that intracellular bacteria
- IFN- γ also tells macrophage to increase production of toxic enzymes (NADPH oxidase and nitric oxide synthetase) to incre production of toxic oxygen metabolites and nitric oxide

1. Effector Th1 cells secrete IFN- γ that activates infected macrophages → increases lysosome-phagosome fusion



- However, that is not v. destructive for 100% of the intracellular bacteria
- Another role of Effector Th1 cell is to “wall off”/ surround the infected macrophages in the tissue and “suffocate” the bacteria
 - Called granuloma
 - Blocks entry of nutrients and O₂ -> starves the intracellular bacteria
 - Kills infected macrophages but dn always kill the bacteria (ie tuberculosis -> TB remains latent for life - we cannot 100% kill these pathogens)
- In summary:

in summary

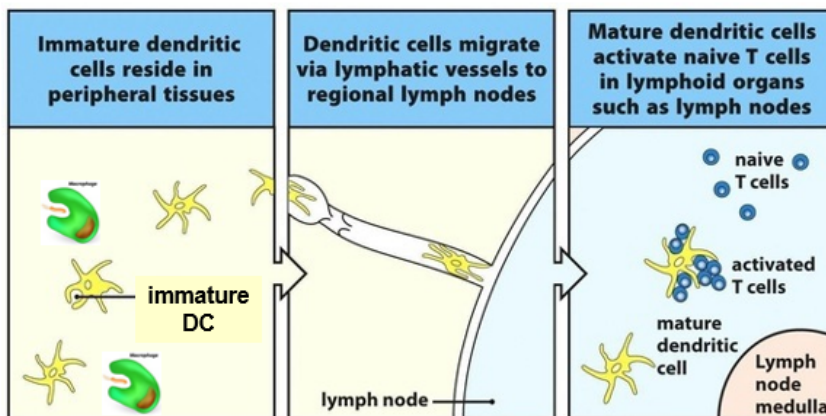


- Mature DC presents peptide that naïve cell binds to
 - MDC secretes polarizing cytokines onto naïve T helper cell telling it to become effector Th1 cell
 Daughter cells become effective Th1 cells exit SLT and enter into infected tissue via inflammation (same as effector CTL)

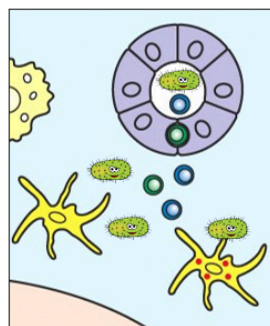
- Once in tissue, it binds to infected macrophage (MHC class 2) and it produces IFN- γ which binds to interferon gamma receptor
- sends activation signal to macrophage to produce more toxic O₂ metabolites + nitric oxide, and increase phagosome-lysosome fusion -> (not that effective tho)
- Aldo forms **granuloma** to hopefully suffocate the infected macrophages (not always effective)

Topic 7-3

- Immature dendritic cells reside in blood & tissues
 - Includes peripheral tissue and secondary lymphoid tissue
- Pathogen can enter a dendritic cell by:
 - 1) **Phagocytosis** (extracellular bacteria)- peptides presented on MHC class 2 molescs
 - 2) **Viral entry or intracellular bacteria** (DC infected) - peptides presented on MHC class 1 molescs
- **Immature DC** “internalize” pathogens in peripheral tissues
- Some DC exit tissues and migrate via **lymphatics** to **lymph node**
- DCs *mature* during migration & become **APC (Antigen-Presenting Cell)** (aka mature DC)
- **Mature DC** enters T cell area of Lymph Node (LN) and present its peptides on MHC molecules to **naïve T cells**



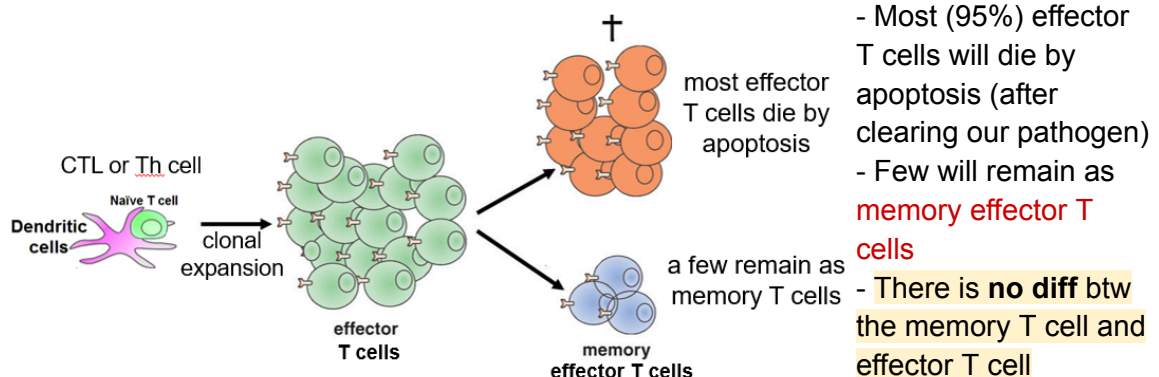
- Some immature DC are resident in SLT (secondary lymphoid tissue) & meet pathogens entering the **spleen** (via blood), **lymph node** (via lymph = pathogens in peripheral tissues), and **MALT** (via mucosa)



- DC “internalize” incoming pathogens.
- Present peptides on MHC molecules to **naïve T cells** circulating through SLT.

- Naive T cell will interact w/ a migrant peripheral DC at the lymph node, or they will interact w/ a resident DC at the spleen, lymph node, or the MALT tissue
- 1 naive T cell circulates thru a lymph node every 12-24hrs
- Macrophages that are present in peripheral tissue *do not* exit the peripheral tissue, unlike DC
- In mucosal areas - say a cut in your mouth
 - Pathogens enter and encounter immature DC under epithelial cell layer
 - Immature DC take up pathogen and some will enter lymphatics and enter regional lymph node
 - Some pathogens that entered can also enter MALT tissues via the **M cell** (ie tonsils in mouth)
 - Naive T cells will be responding to pathogen's peptides in both MALT tissues and lymph nodes
- Fun fact: Of the 800 lymph nodes present in the human body, 300 are in the neck region
 - This is bc most pathogens enter our body via our mouth

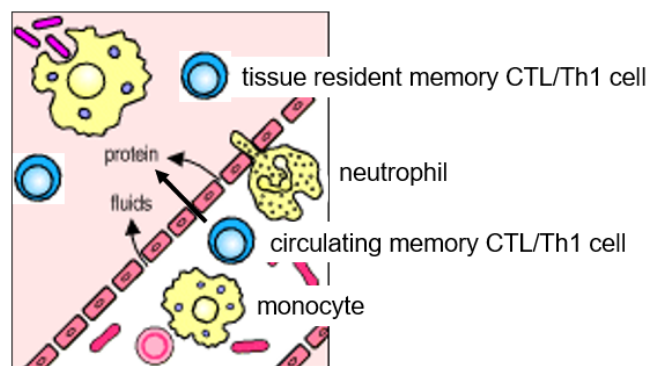
- Recall: T cell immunity always initiated in SLT by mature dendritic cell to naive T cell



- We *do not* know how memory T cells are regulated

Memory T cell

- Response is more effective bc it's faster (we already generated those effector T cells so they can respond ASAP to re-exposure of the identical pathogen)
- Reside in SLTs and peripheral tissues
- Upon re-infection, memory T cells can act alongside innate immunity (neutrophils, monocytes, immature DC)



(next couple slides are review slides)

Topic 8-1

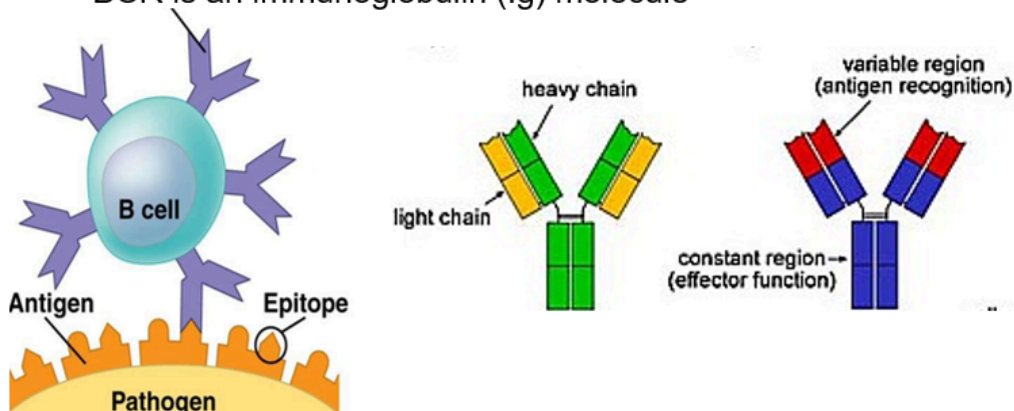
- B cell immunity aka Humoral Immunity

B CELL IMMUNITY LEARNING OBJECTIVES:

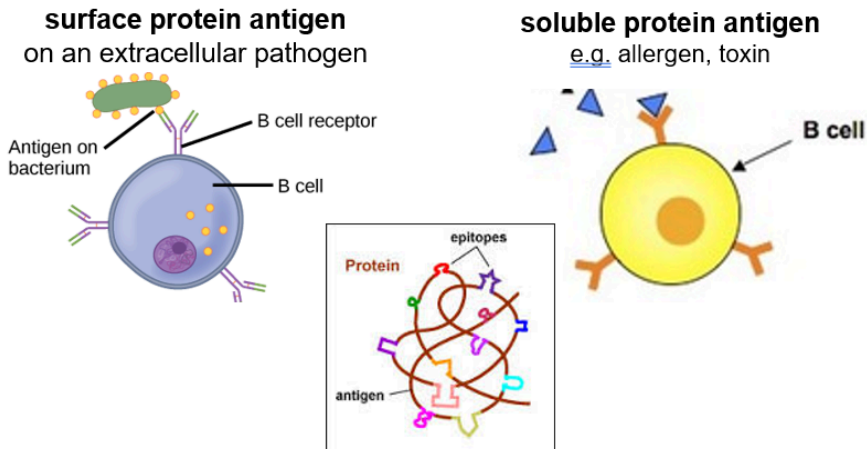
By the end of this section, you should know:

1. The kinetics of antibody production following primary exposure vs. re-exposure to a protein vs. non-protein antigen.
 2. What cells are generated in a primary vs. memory immune response?
 3. How antibodies function.
 4. The process of somatic hypermutation and class switching.
 5. What B cell components provide longterm immune protection?
 6. Passive immunity in a newborn and an adult.
- B cell immunity results in production of **antibodies** to help eliminate **extracellular pathogens** and inactive **soluble proteins** (e.g toxins)
 - Each B cell expresses multiple copies (approx 100,000) of a **B cell receptor (BCR)** with specificity to **one** unique **antigenic epitope**
 - BCR is an **immunoglobulin (Ig)** molecule
 - structure: 2 identical heavy chains & 2 identical light chains; see right image for constant region and variable regions
 - Ig molec expresses 2 identical variable regions - can bind to the same antigenic epitope twice
 - Can bind directly to **antigenic epitope**, unlike T cell receptor
 - TCR can only bind to to one peptide (protein antigen); BCR can bind directly to both a protein and non-protein antigens

BCR is an immunoglobulin (Ig) molecule

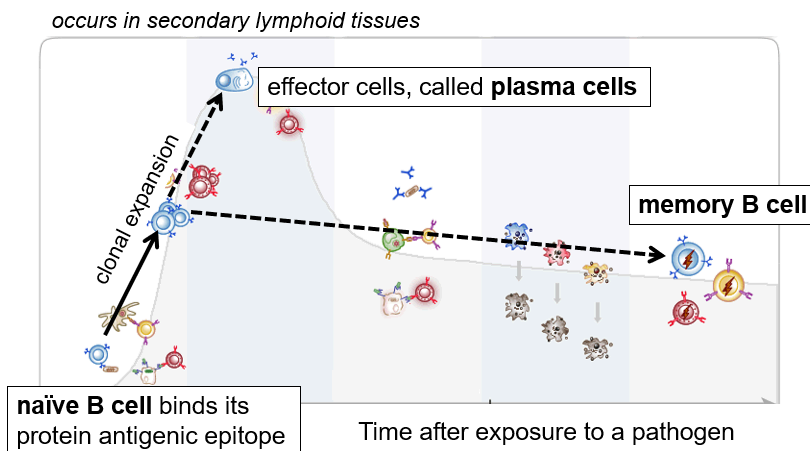


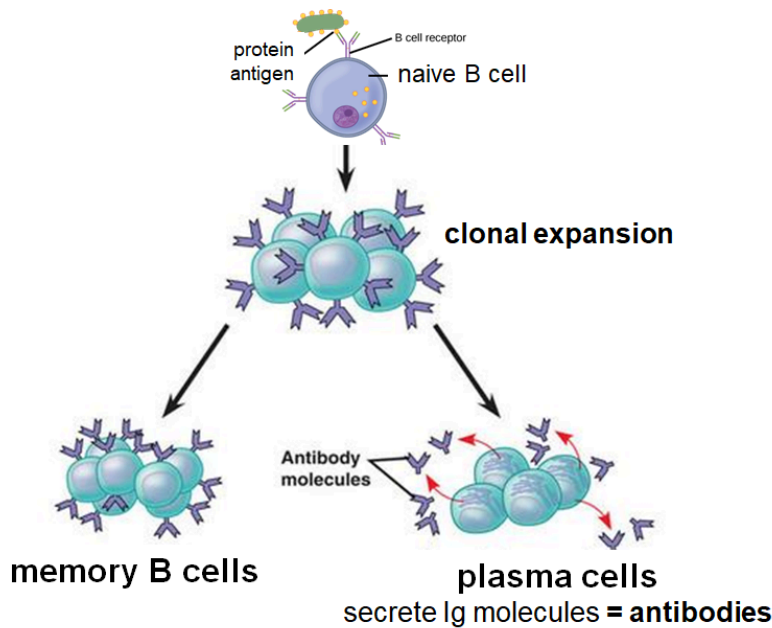
- Most common B cell response is against **protein antigens**



- Can be **surface protein antigen** or **soluble protein antigen**
- Can see that protein has multiple different epitopes on it
- Epitopes are where the variable region of B cell receptor and antibody will bind to

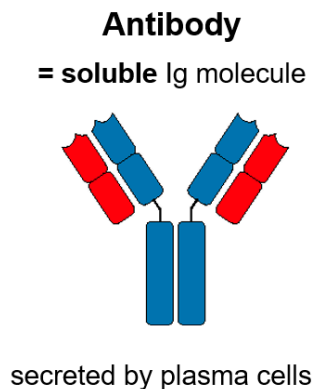
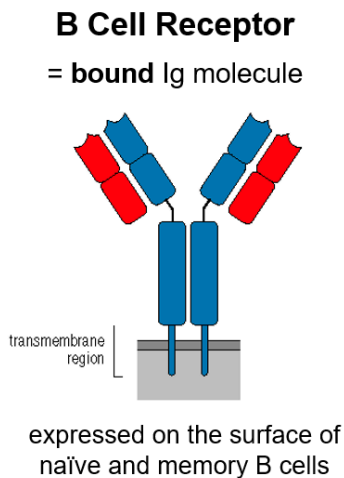
- Reviewing a primary immune response:
 - With B cell immunity, the naive B cell is circulating through the secondary lymphoid tissues -> if it finds its unique **protein antigenic epitope** it will bind directly to it (could be on extracellular pathogen or soluble protein)
 - Naive B cell then undergoes clonal expansion
 - Daughter cells will become either effector cells (aka **plasma cells**) or memory B cells
 - **Different from T cell immunity where all daughter T cells become effector cells**
 - Primary B cell response to a protein antigen generates plasma cells and memory B cells





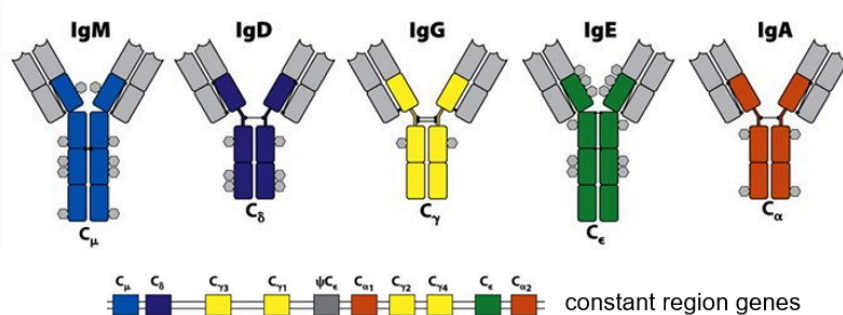
- Daughter cells will become either effector cells (aka **plasma cells**) or memory B cells
- Plasma cells secrete immunoglobulin molecules called **antibodies**
- Secretes approx 2000 antibodies per sec for 3-5 days
- whether daughter cell becomes memory or plasma cell not well understood

- Structure of B cell receptor and antibody



- B cell receptor and antibody are both immunoglobulin molecules
- the difference is that B cell receptor is bound to the surface of B cell (naïve or memory), while antibody is a soluble Ig molecule (secreted by plasma cells)

- 5 classes of immunoglobulin molecules, also called **isotypes**



- molecules have diff gene regions w/in heavy chain constant region

- **hinges** gives the molecule flexibility so it can flex and bind to its **antigenic epitope** at a **higher affinity** than a **ig molecule** can

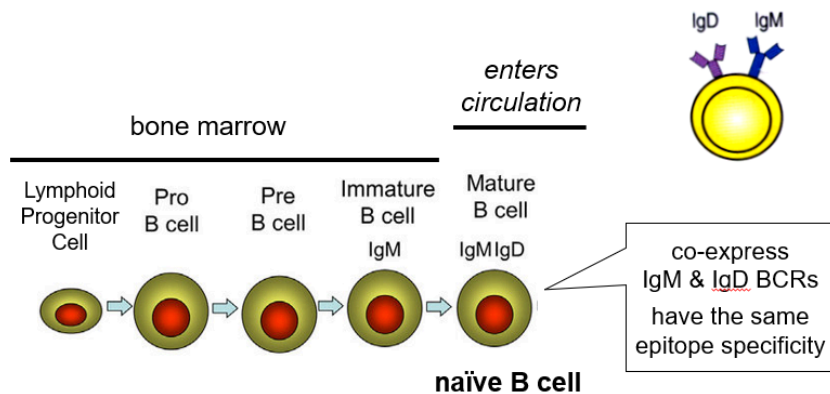
Class is defined by differences in **heavy chain constant region**

- Length
- Distribution of carbohydrates
- Location of hinges (disulphide bonds)



- IgE (see image above) do have a role in immunity against bacteria and viruses - instead it's important against parasites

B Cell Development



- When B cell is developing, it starts as a **hematopoietic stem cell**, then **lymphoid progenitor cell**, then slowly develops into B cell (see image)

- Once B cell enters into circulation, it is a naïve B cell and co-expresses both **IgM & IgB** B cell receptor

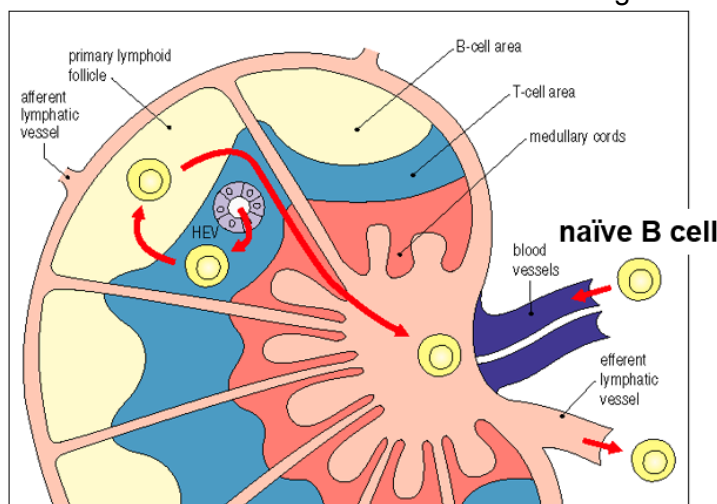
- IgM and IgB receptor have exactly the same variable

region - have exact same specificity for same epitope (recall: one B cell has specificity for one unique antigenic epitope)

- Why B cell expresses both receptors remain unclear

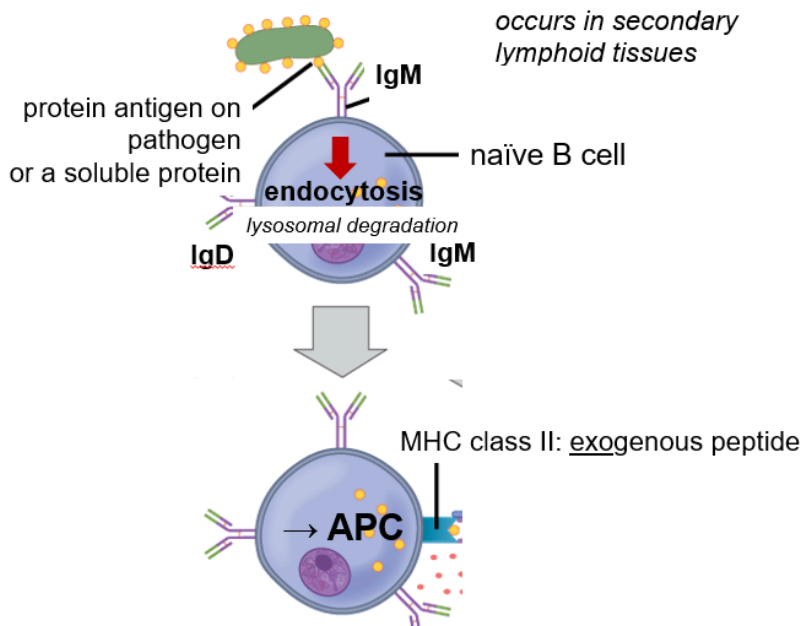
Topic 8-2

- Like T cells, B cells circulate thru SLT
 - If it do bind to its antigen it remains inactive and will re-circulate for a few months. If it do comes into contact with its antigen w/in that time it will undergo apoptosis



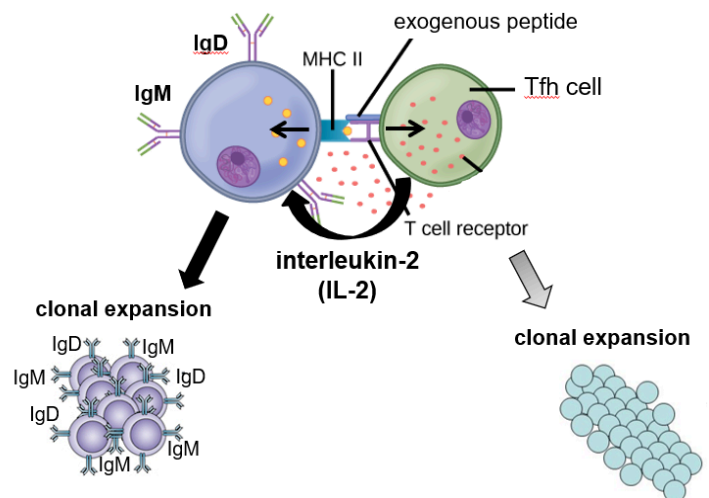
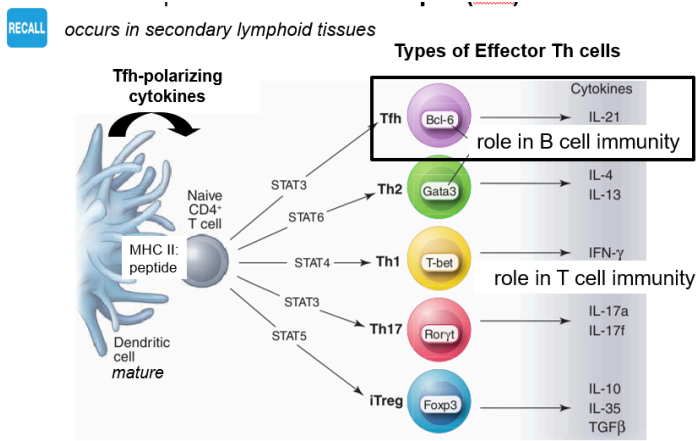
- Recall: Most common B cell responses is against **protein antigens**
 - **Surface protein antigen** on an extracellular pathogen
 - **Soluble protein antigen** → ie: allergen, toxin

- B cell immune responses to a protein antigen



- If naive B cell encounters its specific protein antigen or extracellular pathogen or a soluble protein, that naive B cell will bind it by its variable region and then internalize it via endocytosis
- Recall: endocytosis -> lysosomal degradation so exogenous peptides generated will be presented by that B cell on MHC class 2 molecules
- Recall: B cells are key **antigen presenting cells (APC)**

- B cell immunity against a protein antigen requires “help” from T follicular helper (Tfh) cells

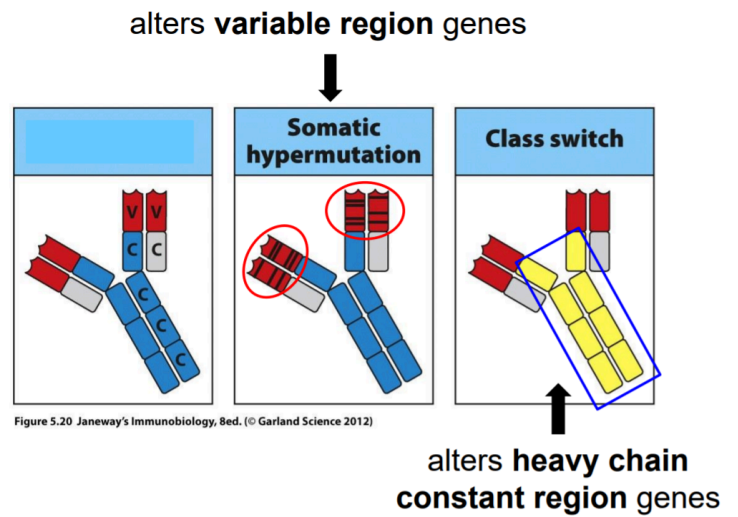
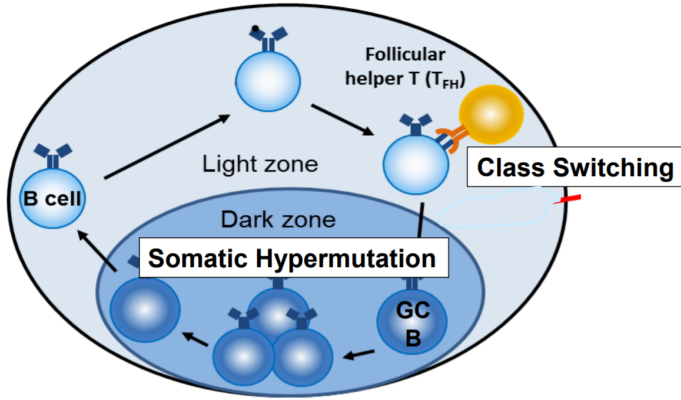


- Tfh cytokines activate *clonal expansion*
 - Tfh has a T cell receptor w/ a variable region specific for that peptide being presented by the B cell
 - Binding interaction activates both the B cell and Tfh cell
 - Tfh cell begins to secrete **cytokine interleukin-2 (IL-2)**
 - IL-2 activates B cell's clonal expansion
 - Tfh will also undergo clonal expansion
- T cell immunity must occur to initiate B cell immunity
- This all happens in SLT

Summary below

Topic 8-3

- 2 GC events that occur are **Class switching** and **Somatic Hypermutation**
 - SM alters the variable regions genes
 - CS alters heavy chain and constant regions



- SHM incre variable region's affinity for its epitope
 - The variable region's specificity remains the same

